

Selective Estimation of the Nonmatch Distribution in `mec_blocking()`

Methodological specification for StatsClaw

May 2026

Scope

This note specifies one targeted change to the blocking-based inverted MEC procedure. The rest of the method remains unchanged: the algorithm works on the blocking candidate-pair space, estimates the match distribution from all blocked pairs, estimates the nonmatch distribution iteratively, uses the inverted density ratio

$$s(\gamma) = \frac{u(\gamma)}{m(\gamma)},$$

and constructs the updated one-to-one match set in increasing order of s .

The proposed change concerns only the sample used to estimate the nonmatch distribution u . Instead of estimating u from the intersection of the current nonmatch set and the initial nonmatch set, estimate u from the most reliable part of the current nonmatch set. The only exception is the first iteration: before any q 's or s 's are available, the full initial nonmatch set should be used.

Motivation

Let Ω_B be the blocked candidate-pair space and $N = |\Omega_B|$. At iteration t , let $M^{(t)}$ be the current one-to-one match set and

$$U^{(t)} = \Omega_B \setminus M^{(t)}$$

be the current nonmatch set.

The current rule based on $U^{(t)} \cap U^{(0)}$ is conservative but strongly anchored to the initialization. If the observed false link rate is very low while the missing match rate is too high, the procedure is likely too conservative. Then $U^{(t)}$ may contain missed true matches. If such match-like pairs are used to estimate u , the estimated nonmatch distribution becomes too close to the match distribution, which may further increase missing matches.

The proposed rule therefore separates two objects:

$$\begin{aligned} U^{(t)} & \text{ current nonmatch class,} \\ S_U^{(t)} \subseteq U^{(t)} & \text{ estimation subset for } u. \end{aligned}$$

Only $S_U^{(t)}$ is used for estimating u . All MEC count and posterior formulas must still use the full $|U^{(t)}|$, not $|S_U^{(t)}|$.

New parameter

Introduce a parameter

$$\lambda \in [0, 1),$$

called, for example, `nonmatch_drop_fraction`. It controls what fraction of the current nonmatch set is removed from the estimation sample after the first distribution-estimation step.

The first iteration is deliberately uncorrected:

$$S_U^{(0)} = U^{(0)}.$$

This is because no previous nonmatch posterior probabilities or inverted-ratio values are available, and the initialization should not be altered by an additional reliability filter.

For iterations $t \geq 1$, define

$$n_{\text{drop}}^{(t)} = \lfloor \lambda |U^{(t)}| \rfloor, \quad n_{\text{keep}}^{(t)} = |U^{(t)}| - n_{\text{drop}}^{(t)}.$$

Thus $\lambda = 0$ uses all current nonmatches to estimate u . Larger λ values estimate u from a cleaner but less representative subset of nonmatches.

The implementation should enforce the minimum sample size needed by the selected model. If the retained subset is too small, reduce $n_{\text{drop}}^{(t)}$ for that iteration or use $S_U^{(t)} = U^{(t)}$, and report this in diagnostics.

Selection of $S_U^{(t)}$

For $t = 0$, no reliability ranking is applied and $S_U^{(0)} = U^{(0)}$. For $t \geq 1$, each pair in $U^{(t)}$ receives a nonmatch-reliability score. Larger values mean that the pair is more reliable as a nonmatch. Use the previous nonmatch posterior $\hat{q}_{ab}^{(t-1)}$, or equivalently the previous inverted ratio $s^{(t-1)}(\gamma_{ab})$. This avoids using the initialization norm a second time.

For $t \geq 1$, sort $U^{(t)}$ by this score in decreasing order. Define $S_U^{(t)}$ as the first $n_{\text{keep}}^{(t)}$ pairs in that order. Equivalently, remove the $n_{\text{drop}}^{(t)}$ least reliable nonmatches from the estimation sample. Ties should be resolved deterministically, for example by pair identifiers.

Modified iteration

At iteration t :

1. Set $U^{(t)} = \Omega_B \setminus M^{(t)}$.
2. If $t = 0$, set $S_U^{(0)} = U^{(0)}$. If $t \geq 1$, select $S_U^{(t)} \subseteq U^{(t)}$ using `nonmatch_drop_fraction` and the previous iteration's reliability scores.
3. Estimate $\hat{u}^{(t)}$ from $S_U^{(t)}$.

4. Compute

$$s^{(t)}(\gamma) = \frac{\widehat{u}^{(t)}(\gamma)}{\widehat{m}(\gamma)},$$

where $\widehat{m}$ is still estimated from all pairs in Ω_B .

5. Compute nonmatch posteriors with the full $|U^{(t)}|$:

$$\widehat{q}_{ab}^{(t)} = \min \left\{ \frac{|U^{(t)}| s^{(t)}(\gamma_{ab})}{|U^{(t)}| \{s^{(t)}(\gamma_{ab}) - 1\} + N}, 1 \right\}.$$

6. Continue with the existing inverted MEC steps: estimate the number of nonmatches, convert it to the number of matches, and build the updated one-to-one match set in increasing order of $s^{(t)}$.

Distributional cases

For the binary MEC variant, estimate the nonmatch agreement probabilities from $S_U^{(t)}$. For example,

$$\widehat{\xi}_k^{(t)} = \frac{1}{|S_U^{(t)}|} \sum_{(a,b) \in S_U^{(t)}} \gamma_{ab}^k,$$

using the same smoothing and boundary handling as in the current implementation.

For the continuous parametric variant, estimate the hurdle-Gamma nonmatch parameters from $S_U^{(t)}$: the mass at zero from the proportion of zeros and the Gamma parameters from positive comparison values. Existing safeguards for zero counts, empty positive parts, or degenerate samples should be applied to $S_U^{(t)}$.

Expected effect and validation

The parameter λ controls the purity-representativeness tradeoff. Larger values remove more match-like pairs from nonmatch estimation, which may reduce missing matches when the method is overly conservative. Too large a value may estimate u only from easy nonmatches and can increase false links. Validation should therefore compare several λ values and focus on the FLR–MMR tradeoff.

The implementation should preserve these invariants:

- $S_U^{(0)} = U^{(0)}$, and `nonmatch_drop_fraction` is applied only for $t \geq 1$;
- $S_U^{(t)} \subseteq U^{(t)}$ and $U^{(t)} = \Omega_B \setminus M^{(t)}$;
- posterior calculations use $|U^{(t)}|$, not $|S_U^{(t)}|$;
- the final match set remains one-to-one;
- initially removed pairs are not frozen and may re-enter $M^{(t)}$ later.